

Information warfare (IW) is the battlespace use and management of information and communication technology (ICT) in pursuit of a competitive advantage over an opponent. It is different from cyberwarfare that attacks computers, software, and command control systems. Information warfare is the manipulation of information trusted by a target without the target's awareness so that the target will make decisions against their interest but in the interest of the one conducting information warfare. As a result, it is not clear when information warfare begins, ends, and how strong or destructive it is.

Information warfare may involve the collection of tactical information, assurance(s) that one's information is valid, spreading of propaganda or disinformation to demoralize or manipulate the enemy and the public, undermining the quality of the opposing force's information, and denial of information-collection opportunities to opposing forces. Information warfare is closely linked to psychological warfare.

Overview

Information warfare has been described as "the use of information to achieve our national objectives." According to NATO, "Information war is an operation conducted in order to gain an information advantage over the opponent." The term has existed at least since 1970, when Dale Minor, a reporter, published *The Information War* a book about propaganda and news manipulation during the Vietnam War era., although its current usage as a military capability dates from 1976, when Thomas P. Rona, then a Boeing Company engineer, referred to "information war" in a company monograph prepared for the Office of Net Assessment.

Information warfare can take many forms:

Television, internet and radio transmission(s) can be jammed to disrupt communications, or hijacked for a disinformation campaign.

Logistics networks can be disabled.

Enemy communications networks can be disabled or spoofed, especially online social communities in modern days.

Stock exchange transactions can be sabotaged, either with electronic intervention, by leaking sensitive information or by placing disinformation.

The use of drones and other surveillance robots or webcams.

Communication management

Synthetic media

The organized use of social media and other online content-generation platforms can be used to influence public perceptions.

The United States Air Force has had Information Warfare Squadrons since the 1980s. In fact, the official mission of the U.S. Air Force is now "To fly, fight and win... in air, space and cyberspace", with the latter referring to its information warfare role.

As the U.S. Air Force often risks aircraft and aircrews to attack strategic enemy communications targets, remotely disabling such targets using software and other means can provide a safer alternative. In addition, disabling such networks electronically (instead of explosively) also allows them to be quickly re-enabled after the enemy territory is occupied. Similarly, counter-information warfare units are employed to deny such capability to the enemy. The first application of these techniques was used against Iraqi communications networks in the Gulf War.

Also during the Gulf War, Dutch hackers allegedly stole information about U.S. troop movements from U.S. Defense Department computers and tried to sell it to the Iraqis, who thought it was a hoax and turned it down. In January 1999, U.S. Air Intelligence computers were hit by a coordinated attack (Moonlight Maze), part of which came from a Russian mainframe. This could not be confirmed as a Russian cyber attack due to non-attribution – the principle that online identity may not serve as proof of real-world identity.

Some militaries are now employing the use of iPhones to upload data and information

gathered by drones in the same area.

Notable examples

Chinese information warfare

Russo-Ukrainian War

In 2022, the Armed Forces of Ukraine took advantage of deficiencies in Russian communications by allowing them to piggyback on Ukrainian networks, connect, and communicate. Ukrainian forces then eavesdrop and cut off Russian communications at a crucial part of the conversation.

To build support before it invaded Ukraine, Russia perpetuated a narrative that claimed the Ukrainian government was committing violence against its own Russian speaking population. By publishing large amounts of disinformation on the internet, the alternate narrative was picked up in search results, such as Google News.

Russian interference in foreign elections

Russian interference in foreign elections, most notably the Russian interference in the 2016 United States elections, has been described as information warfare. According to Microsoft, Russia also interfered in the 2024 US presidential elections. NBC also reported Russia conducting disinformation campaigns in the 2024 US elections against then US president, Joe Biden.

Russia vs West

Research suggests that Russia and the West are also engaged in an information war. For instance, Russia believes that the West is undermining its leader through the encouragement of overthrowing authoritarian regimes and liberal values. In response, Russia promotes the anti-liberal sentiments, including racism, antisemitism, homophobia, and misogyny. Russia

has sought to promote the idea that the American democratic state is failing.

Russia, China and pro-Palestinian protests

The Telegraph reported in 2024 that China and Russia were promoting pro-Palestinian influencers in order to manipulate British public opinion in favor of Russian and Chinese interests. NBC reported that Russia was using different tools to cause division within the US, by delegitimizing US police operations against Pro Palestinian protests and by pivoting public conversation from the Russian invasion in Ukraine to the Israeli-Palestinian conflict. Russian media activity increased by 400% in the weeks after Hamas' Oct. 7 attack on Israel.

United States COVID-19 disinformation campaign

According to a report by Reuters, the United States ran a propaganda campaign to spread disinformation about the Sinovac Chinese COVID-19 vaccine, including using fake social media accounts to spread the disinformation that the Sinovac vaccine contained pork-derived ingredients and was therefore haram under Islamic law. The campaign was described as "payback" for COVID-19 disinformation by China directed against the U.S. The campaign ran from 2020 to mid-2021, primarily targeting people in the Philippines and used a social media hashtag for "China is the virus" in Tagalog. The primary contractor for the U.S. military on the project was General Dynamics IT, which received \$493 million for its role.

Israeli information warfare

Israel has used a variety of methods to influence public opinion in recent years, from advertising and paying influencers to creating networks of websites to influence the responses of AI chatbots. According to a report by Drop Site News, Israel has launched a digital campaign to influence the response of AI systems to the Gaza war, in the form of a \$46.5 million contract with Clock Tower X, a company headed by Brad Parscale, a former

Trump campaign manager. The company has published hundreds of articles on a network of websites that support Israel's positions on the Gaza war, its military operations, and US-Israeli relations. The aim is to get into the data used by AI systems such as ChatGPT, Claude, and Perplexity, a method called "LLM poisoning." In an experiment conducted by Drop Site, a chatbot was asked whether increasing US military cooperation with Israel would be beneficial; The chatbot answered "yes," and the first source it cited was a website that Parscale had created for the Israeli government.

Legal and ethical concerns

While information warfare has yielded many advances in the types of attack that a government can make, it has also raised concerns about the moral and legal ambiguities surrounding this particularly new form of war. Traditionally, wars have been analyzed by moral scholars according to just war theory. However, with Information Warfare, Just War Theory fails because the theory is based on the traditional conception of war. Information Warfare has three main issues surrounding it compared to traditional warfare:

The risk for the party or nation initiating the cyberattack is substantially lower than the risk for a party or nation initiating a traditional attack. This makes it easier for governments, as well as potential terrorist or criminal organizations, to make these attacks more frequently than they could with traditional war.

Information communication technologies (ICT) are so immersed in the modern world that a very wide range of technologies are at risk of a cyberattack. Specifically, civilian technologies can be targeted for cyberattacks and attacks can even potentially be launched through civilian computers or websites. As such, it is harder to enforce control of civilian infrastructures than a physical space. Attempting to do so would also raise many ethical concerns about the right to privacy, making defending against such attacks even tougher.

The mass-integration of ICT into our system of war makes it much harder to assess accountability for situations that may arise when using robotic and/or cyber attacks. For robotic weapons and automated systems, it's becoming increasingly hard to determine who is responsible for any particular event that happens. This issue is exacerbated in the case of cyberattacks, as sometimes it is virtually impossible to trace who initiated the attack in the first place.

Recently, legal concerns have arisen centered on these issues, specifically the issue of the right to privacy in the United States of America. Lt. General Keith B. Alexander, who served as the head of Cyber Command under President Barack Obama, noted that there was a "mismatch between our technical capabilities to conduct operations and the governing laws and policies" when writing to the Senate Armed Services Committee. A key point of concern was the targeting of civilian institutions for cyberattacks, to which the general promised to try to maintain a mindset similar to that of traditional war, in which they will seek to limit the impact on civilians.

See also

Group specific:

US specific:

Notes

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